IPO Portfolio Optimization with GRU-Based Allocation

STAT 4830 — Spring 2026

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Outline

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3. Model Architecture
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Problem Setup & Objective

Task: Learn daily portfolio weights between two assets

1. **Market Index** — 82% SPY + 18% DIA
2. **IPO Index** — Market-cap weighted IPO portfolio (held 180 trading days post-IPO)

Model outputs

$$\mathbf{w}_t = (w_t^{\text{mkt}},\ w_t^{\text{ipo}}) \in \Delta^1, \qquad w_i \geq 0,\ \sum w_i = 1$$

via a **GRU** mapping rolling return windows to allocations.

Optimization Objective

$$\mathcal{L} = -\bar{r}_p + \lambda_{\text{cvar}} L_{\text{CVaR}} + \lambda_{\text{vol}} \text{Var}(r_p) + \lambda_{\text{vol_ex}} \max(0, \hat{\sigma}_{\text{ann}} - \sigma^*)$$
$$+\lambda_{\text{turn}} \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1 + \lambda_{\text{path}} \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_2^2 + \lambda_{\text{div}} \max(0, w_{\min}^* - \min_i w_i)$$

Mean return maximization with explicit penalties for tail risk, volatility targeting, turnover control, path stability, and diversification. Softmax ensures feasibility without projection.

Why IPOs?

IPOs exhibit distinct return patterns:

- **Higher volatility**: 3–4$\times$ the market
- **Mean reversion** in the first 6 months
- **Information asymmetry** between institutional and retail investors

A *systematic allocator* that adjusts IPO exposure based on recent history can improve risk-adjusted returns compared to static benchmarks.

Use case: Retail and institutional investors use the model output as a *tilt signal* (how much IPO vs. market exposure to hold).

Success Metrics & Constraints

Success Metrics

Metric	Target
Sharpe Ratio	> 1.5
Max Drawdown	$< 15\%$
Ann. Return	$>$ Market
Turnover	Low

Constraints

- Long-only, fully invested
  $w_i \geq 0, \ \sum w_i = 1$
- No shorting or leverage
- Daily rebalancing

IPO Index Construction

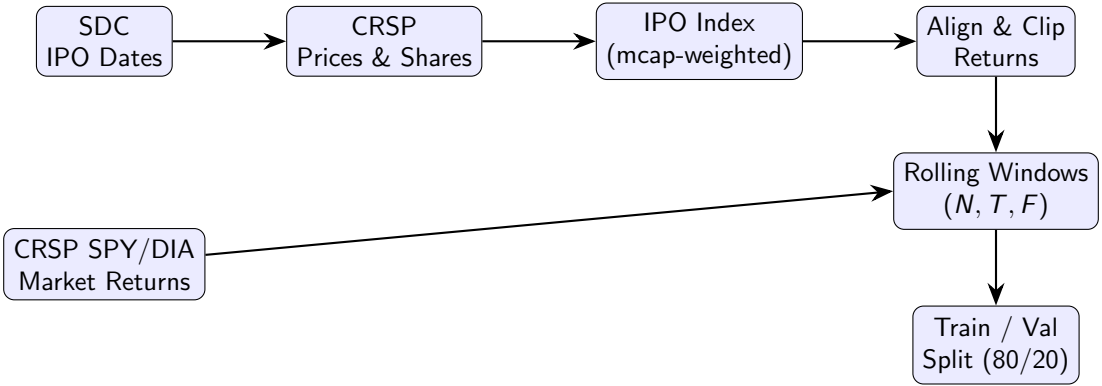
1. Query SDC for all IPOs (`ipo = 'Yes'`, `ipodate IS NOT NULL`)
2. Pull daily prices & shares outstanding from CRSP (`crsp.dsf`)
3. For each trading day t, identify tickers within 180 days of their IPO date
4. Compute **market-cap weighted** return:

$$r_t^{\text{ipo}} = \sum_{i \in \mathcal{A}_t} \frac{\text{mcap}_{i,t}}{\sum_{j \in \mathcal{A}_t} \text{mcap}_{j,t}} \cdot r_{i,t}$$

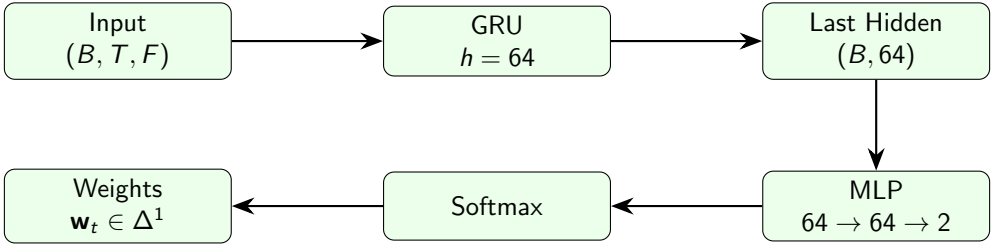
where $\mathcal{A}_t = \{i : 0 \leq \text{age}_{i,t} < 180\}$ and $\text{mcap}_{i,t} = |P_{i,t}| \times \text{shrout}_{i,t}$.

Market returns: $r_t^{\text{mkt}} = 0.82 \cdot r_t^{\text{SPY}} + 0.18 \cdot r_t^{\text{DIA}}$

Data Processing Pipeline



GRU-Based Allocator



- **Input**: Rolling window of $T = 126$ days $\times$ F features (market ret, IPO ret, rolling vol, VIX)
- **GRU** captures temporal dependence in returns
- **Softmax** output enforces simplex constraint (long-only, fully invested)
- Differentiable end-to-end $\Rightarrow$ gradient-based optimization

Training Pipeline

Training Loop

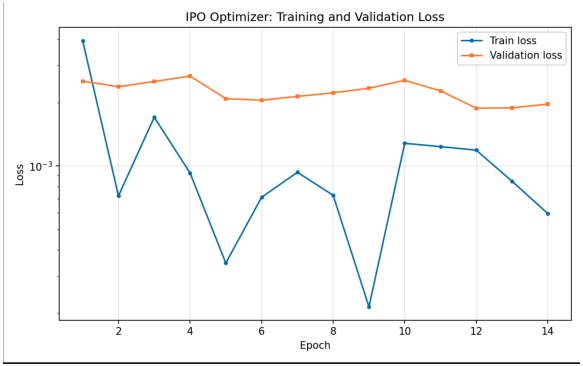
1. Build rolling windows (X, R)
2. Time-series split: last 20% $\rightarrow$ validation
3. Mini-batch SGD with Adam (lr $= 10^{-3}$, weight decay 10^{-5})
4. Gradient clipping (max norm 1.0)
5. Early stopping (patience = 10 epochs)
6. Restore best checkpoint (min val loss)

Key Hyperparameters

Parameter	Value
`window_len`	126
`hidden_size`	64
`batch_size`	32
`val_frac`	0.2
$\lambda_{\text{vol_excess}}$	1.0
σ^* (target vol)	25%
λ_{cvar}	0.5
λ_{vol}	0.5

Hyperparameter tuning: 32-config grid search over window length, vol penalties, CVaR, etc.

Training Convergence



- Training converges in **11–14 epochs** with early stopping
- Weights satisfy simplex constraint; no NaN/Inf

Validation Results (2020–2024)

Strategy	Total	Ann. Ret.	Ann. Vol	Sharpe	Max DD
Model	**34.39%**	**39.44%**	13.52%	**2.53**	**-7.66%**
Market only	17.26%	19.62%	12.22%	1.53	-7.89%
IPO only	166.59%	201.35%	30.68%	3.75	-10.08%
Equal 50/50	78.16%	91.50%	19.26%	3.47	-7.20%

What the Model Actually Learned

- Average allocation: $w^* \approx$ (0.84 market, 0.16 IPO)
- Never exceeds 20% IPO exposure
- Extremely low turnover ($\sim 10^{-5}$ daily)
- Behavior resembles a *disciplined core + satellite tilt*

Interpretation:
The model does *not* chase the high raw returns of IPO-only. It extracts a controlled IPO premium while preserving market-like drawdowns. Risk penalty dominates extreme tilting, producing stable, institutional-style behavior.

Strengths

1. **Rigorous data pipeline**: Institutional WRDS data (SDC + CRSP) with split-adjusted prices and shares. 770K rows, 1,136 tickers.
2. **Differentiable, modular loss**: Cleanly decomposed objective with configurable λ's. Supports interpretability and systematic tuning.
3. **Evidence of validation**: 32-config grid search; early stopping; baseline comparisons provide context.
4. **End-to-end differentiability**: Softmax enforces constraints without projection. All components are gradient-friendly.

Limitations

1. **No true out-of-sample test**: All metrics on validation set (last 20% of 2020–2024). No held-out 2025 test period.
2. **Near-constant weights**: $\approx 84/16$ split every day. GRU may be underfitting temporal variation, or penalties suppress turnover.
3. **Survivorship bias**: IPO index excludes delisted stocks still present in CRSP.
4. **Regime dependence**: 2020–2024 includes strong IPO performance (COVID recovery, IPO boom). Future regimes may differ.

Next Steps

Immediate

- True out-of-sample test (reserve 2025 or last 60 days)
- Improve our model architecture to work on our validation loss
- Expand dataset to include a wider range of dates and stocks

Technical

- Include delisted IPOs to address survivorship bias
- Sensitivity analysis on hyperparameters
- Utilize a wavelet in pre-processing

Alternative Approaches

- Reinforcement learning (PPO/SAC)
- Online learning: incremental model updates